# Assignment 3

## Problem 3.1

$3 \times 3$ warehouse

Propositional symbols:

$D_{i,j}$ : Damaged floor at $i,j$

$F_{i,j}$ : Forklift at $i,j$

$P_{i,j}$ : Package at $i,j$

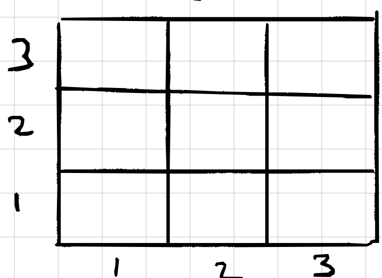
$C_{i,j}$ : Creaking perceived at $i,j$

$R_{i,j}$ : Rumbling perceived at $i,j$

$S_{i,j}$ : Square $(i,j)$ is safe

1) Encode Physics

- Creaking at a location iff damaged floor in an adjacent location.




$C_{1,1} \Leftrightarrow D_{1,2} \vee D_{2,1}$

~~$C_{1,2} \Leftrightarrow D_{1,3} \vee D_{2,2}$ because $S_{1,1}$~~

$C_{1,3} \Leftrightarrow D_{1,2} \vee D_{2,3}$

~~$C_{2,1} \Leftrightarrow D_{2,2} \vee D_{3,1}$ because $S_{1,1}$~~

$C_{2,2} \Leftrightarrow D_{1,2} \vee D_{2,3} \vee D_{2,1} \vee D_{3,2}$

$C_{2,3} \Leftrightarrow D_{2,2} \vee D_{1,3} \vee D_{3,3}$

$C_{3,1} \Leftrightarrow D_{2,1} \vee D_{3,3}$

$C_{3,2} \Leftrightarrow D_{3,3} \vee D_{2,2} \vee D_{3,1}$

$C_{3,3} \Leftrightarrow D_{2,3} \vee D_{3,2}$

$C_{1,2} \Leftrightarrow D_{1,1} \vee D_{1,3} \vee D_{2,2}$

$C_{2,1} \Leftrightarrow D_{1,1} \vee D_{3,1} \vee D_{2,2}$

- Rumbling at a location iff forklift in an adjacent square

Rewrite equations for damage floor because same conditions apply.

$R_{1,1} \iff F_{1,2} \vee F_{2,1}$
$R_{1,2} \iff F_{1,3} \vee F_{2,2} \vee F_{1,1}$
$R_{1,3} \iff F_{1,2} \vee F_{2,3}$
$R_{2,1} \iff F_{2,2} \vee F_{3,1} \vee F_{1,1}$
$R_{2,2} \iff F_{1,2} \vee F_{2,3} \vee F_{2,1} \vee F_{3,2}$
$R_{2,3} \iff F_{2,2} \vee F_{1,3} \vee F_{3,3}$
$R_{3,1} \iff F_{2,1} \vee F_{3,3}$
$R_{3,2} \iff F_{3,3} \vee F_{2,2} \vee F_{3,1}$
$R_{3,3} \iff F_{2,3} \vee F_{3,2}$

- A square is safe iff it has no damaged floor and no forklift.

$$\left.\begin{aligned} S_{i,j} &\iff \neg D_{i,j} \wedge \neg F_{i,j} \\ &\iff \neg (D_{i,j} \vee F_{i,j}) \end{aligned}\right\} \text{for each } (i,j) \in \{1,2,3\}^2$$

Total: 27 propositional sentences encode the physics
27 Initial knowledge

- Starting square (1, 1) is safe.

$$S_{1,1} \Rightarrow \neg D_{1,1} \wedge \neg F_{1,1}$$
$$\Rightarrow \neg (D_{1,1} \vee F_{1,1})$$

- At least one square is damaged

$D_{1,1} \vee D_{2,1} \vee D_{3,1} \vee D_{1,2} \vee D_{2,2} \vee D_{2,3} \vee D_{3,3}$

- At least one square has a forklift

$F_{1,1} \vee F_{2,1} \vee F_{3,1} \vee F_{1,2} \vee F_{2,2} \vee F_{2,3} \vee F_{3,3}$

Total initial knowledge: $1 + 1 + 1 = 3$ sentences

3) Scenario reasoning:

- Robot at (1, 1) and perceives no rumbling and no creaking.

$$\neg C_{1,1}, \quad \neg R_{1,1}$$

- logical inference to determine which squares are provably safe.

| | | | |
|---|---|---|---|
| 3 | | | |
| 2 | S | ? | |
| 1 | $S_{1,1}$ $C \vee R \bot$ | S | ? |
| | 1 | 2 | 3 |

If $C_{1,1} \vee R_{1,1} = \bot$

$$D_{1,2} \vee D_{2,1} \vee F_{1,2} \vee F_{2,1} = \bot$$

$$\neg (D_{1,2} \vee D_{2,1} \vee F_{1,2} \vee F_{2,1})$$

$\neg D_{1,2} \wedge \neg F_{1,2} \Rightarrow S_{1,2}$

$\neg D_{2,1} \wedge \neg F_{2,1} \Rightarrow S_{2,1}$

or

$$\begin{cases} \neg (D_{1,2} \vee F_{1,2}) \Rightarrow S_{1,2} \\ \neg (D_{2,1} \vee F_{2,1}) \Rightarrow S_{2,1} \end{cases}$$

4) Exploration: Robot moves to (2, 1) and perceives creaking but no rumbling.

New percepts $\cancel{\begin{cases} C_{2,1} = T \\ R_{2,1} = \bot \end{cases}}$ $\quad C_{2,1}, \quad \neg R_{2,1}$

$$\begin{cases} F_{2,2} \vee F_{2,3} = \bot \\ D_{2,2} \vee D_{2,3} = T \end{cases}$$

still unknown to be safe but no forklift

Known safe squares $\begin{cases} S_{1,1} \\ S_{2,1} \\ S_{1,2} \end{cases}$